

MOHAMED HANY REDA

Technical IT Project Manager | Software Engineer | AI & Digital Products

Reinforcement Learning · Simulation & Digital Twins · Autonomous Systems · Explainable AI · Production Software Architecture

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PROFESSIONAL SUMMARY

Software engineer and technical project lead who builds complete intelligent systems rather than isolated models. My work spans deterministic simulation and digital twins, reinforcement-learning decision systems, explainable computer vision, and multi-tenant enterprise architecture — each delivered as a working product with a data layer, an interface, and automated tests. I connect business goals, product requirements and engineering execution, and my engineering background lets me understand technical dependencies, identify delivery risks early, and make project decisions with real technical context. Working proficiency in English and Russian alongside native Arabic supports collaboration across international teams.

CORE COMPETENCIES

Technical Project Planning & Roadmapping · Requirements Analysis · Scope & Change Management · Risk Identification & Mitigation · Stakeholder Communication · Cross-Functional Coordination · Technical Decision Support · Product & Business Alignment · System Architecture · Reproducible Engineering · Explainability & Safety by Design · Multilingual Collaboration

TECHNICAL SKILLS

Languages	Python, Dart, C++, C#, SQL, JavaScript
AI / ML	PyTorch, EfficientNet, Computer Vision, Grad-CAM, OpenCV, Scikit-learn, Albumentations, Explainable AI
Reinforcement Learning	PPO, SAC, A2C, Multi-Agent RL, CTDE, Safe-RL, LSTM policies, Self-Play, Domain Randomization
Simulation	Discrete-Event Simulation, Entity-Component-System, Deterministic Replay, Procedural Generation, Sensor Simulation, Operations Research (Erlang C, M/M/c, Little's Law)
Backend	FastAPI, PostgreSQL, SQLAlchemy, Redis, SQLite, REST APIs, Clean Architecture, Multi-Tenant Systems, RBAC
Visualization	Three.js, Pygame, PySide6, Streamlit, Flutter, Unreal Engine 5, Firebase
Engineering	Docker, Git, Pytest, CI, Ruff, MyPy, ONNX

KEY PROJECTS

High-Throughput Sorting-Center Digital Twin

Python · PyTorch · PPO · Multi-Agent RL · LSTM · Discrete-Event Simulation · Three.js — [repository](#)

- Built a logistics digital twin of a 400-lane sorting centre on a custom deterministic discrete-event engine with an Entity-Component-System architecture, making every run reproducible from a single seed.
- Modelled ~96,400 items/hour against a 100,000 items/hour design target, validated against classical queueing-theory baselines.
- Implemented PPO-based parcel routing, multi-agent coordination using centralised training with decentralised execution (CTDE), and an LSTM predictive-maintenance model.
- Added a fault-injection Chaos Agent, time-machine replay with scenario branching, a 2D operations centre and a browser-based 3D twin.
- 530+ automated tests documented in the repository, with Ruff and MyPy in the toolchain.

Strict Drone Safe-RL — Autonomous Control Research Platform

Python · PyTorch · PPO · LSTM · Gymnasium · ONNX · Docker — [repository](#)

- Trained drone-control policies under wind, sensor drift, control latency, motor degradation and GPS-denied navigation.
- PPO with an LSTM actor-critic under Automatic Domain Randomization, plus a Rapid Motor Adaptation-inspired module for online adaptation to changing dynamics.
- Simulated IMU, VIO and LiDAR sensing; five-stage fault-injection curriculum; TrainingGuardian stability monitoring.
- Hard safety limits and emergency fail-safe design, with an ONNX export path for edge deployment.

Simulation and research platform. No real-world autonomous flight validation is claimed.

Fracture Detection AI — Explainable Medical Imaging Product

Python · PyTorch · EfficientNet-B0 · Grad-CAM · Streamlit · SQLite · ReportLab — [repository](#)

- End-to-end computer-vision product for bone-fracture detection on the 4,083-image FracAtlas X-ray dataset.
- Measured results: 90.21% accuracy, 89.31% ROC AUC, 76.67% precision, 63.89% recall, 69.70% F1.
- Grad-CAM overlays make every prediction inspectable; shipped as a Streamlit application and Telegram bot with automated PDF reporting and local SQLite prediction history.
- Trilingual interface (English, Arabic, Russian); 21/21 documented tests passing.

Research and portfolio software. Not a certified medical device.

HMS — Multi-Tenant Enterprise ERP & AI Platform

Python · FastAPI · PostgreSQL · SQLAlchemy · Redis · React · Docker — [repository](#)

- Multi-tenant ERP covering inventory, warehousing, procurement, sales/CRM, finance, maintenance and HR in one Clean Architecture codebase.
- Enforced tenant isolation with PostgreSQL Row-Level Security at the database layer rather than trusting application code.
- Saudi ZATCA Phase 1 e-invoicing compliance, RBAC and audit logging, full Arabic/English RTL support.
- Offline rule-based advisor that runs without any external LLM, with provider-agnostic LLM agents as an optional layer.

AI Evolution Racing Lab — Explainable RL Research

Python · PyTorch · Stable-Baselines3 · PPO/SAC/A2C · Gymnasium — [repository](#)

- Self-evolving simulation where procedurally generated worlds and adaptive difficulty memory drive RL driver training across generations.
- Explainability layer combining policy-entropy confidence, input-gradient saliency and neural activation visualisation.
- AI tournaments, evolution timeline and automated research reports.

Multi-Agent Autonomous Simulation Platform

Python · PyTorch · Multi-Agent RL · PySide6 · SQLite — [repository](#)

- Multi-agent self-play across navigator, observer and interceptor roles in procedurally generated environments with dynamic weather and day/night cycles.
- Decision-probability and value-estimate logging with SQLite episodic memory, surfaced through a real-time PySide6 monitoring dashboard.

PROFESSIONAL EXPERIENCE

Freelance Software & AI Engineer | Self-Employed

2022 — Present

- Design and deliver complete software and AI products end to end — requirements, architecture, implementation and deployment.
- Build reinforcement-learning and simulation systems, and integrate computer-vision models into production-facing tools.
- Design backend systems and relational databases, and integrate internal and third-party REST APIs.
- Act as the technical point of contact for clients, translating business requirements into actionable engineering tasks.

Software Developer | HMS Medical Equipment Company

- Developed the company's web presence and ERP/AI platform work, translating business requirements with stakeholders directly.
- Built responsive interfaces with consistent performance across devices, and optimised page performance.

Software Engineer | Enterprise Inventory Management System

- Designed and built a complete inventory management solution deployed over the company's internal network.
- Implemented authentication and role-based permissions across admin and staff users.
- Built an admin dashboard covering inventory management and operational reporting.
- Designed the SQLite schema and the internal networking layer connecting client workstations.

Android Game Developer | Independent / Google Play

- Designed, developed and published 3 Android games on Google Play, each taken from concept through store release.
- Achieved more than 5,000 combined downloads; ran Google Ads campaigns to support visibility and acquisition.

EDUCATION

Bachelor of Computer Engineering

Tomsk State University of Control Systems and Radioelectronics (TUSUR) · Tomsk, Russia · Expected graduation: 2028

Relevant coursework: Algorithms, Data Structures, Operating Systems, Software Engineering, Artificial Intelligence, Computer Vision, Machine Learning, Networking, Databases.

CERTIFICATIONS & RECOGNITION

- IBM Professional Course — Certificate of Appreciation

- Resala Charity Organization — Certificate of Appreciation, Media Development Manager
- Tomsk State University (TUSUR) — Recognition for contributing to an educational Virtual Reality (VR) system

LANGUAGES

Arabic — Native • **English** — Professional Working Proficiency • **Russian** — Professional Working Proficiency • **German** — Basic